Iron-rich Amazonian lateritic lake sediments harbor diverse microbial communities with biogeochemical potential relevant to carbon and methane cycling

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Abstract

Amazonian lateritic lakes developed on ferruginous canga are seasonally variable, metal-rich systems whose sediment microbiomes remain poorly characterized. We used shotgun metagenomics to investigate microbial communities in sediments from Amendoim, Violão, Três Irmãs and Três Irmãs Adjacent lakes during dry and rainy periods. Coding-sequence taxonomic profiles revealed diverse bacterial and archaeal assemblages and a large unclassified fraction, indicating substantial underexplored diversity. Lake- and season-associated contrasts involved methanogenic, ammonia-oxidizing and anaerobic sediment lineages. Non-metric multidimensional scaling showed partial community overlap, whereas an exploratory, non-significant redundancy analysis placed genus-level variation along loss-on-ignition, aluminium, silica, sulfur and trace-metal gradients. Functional reconstruction identified genetic potential for carbon fixation, methane metabolism, nitrogen and sulfur cycling, photosynthesis, anaerobic respiration and iron metabolism. A curated Kyoto Encyclopedia of Genes and Genomes orthology framework detected 171 of 195 biogeochemical markers and 132 iron-associated markers. Descriptive cross-study contrasts distinguished Amazonian canga-lake profiles from external iron-rich records, but were not treated as inferential tests. We recovered 50 non-redundant metagenome-assembled genomes spanning medium- to high-quality bins, including lineages related to Acidobacteria, Dehalococcoidia, Nitrospirales, Burkholderiales, Bathyarchaeia, Thermoplasmatota and Methanoperedens. These results establish a genome-resolved iron metagenomic atlas for tropical lateritic-lake sediments and a basis for testing how seasonal hydrology and ferruginous geochemistry shape microbial biogeochemical functions.

Introduction

The Serra dos Carajás in southeastern Amazonia is a protected iron-rich landscape where lateritic duricrust, locally termed canga, overlies weathered banded iron formations and high-grade hematite bodies [1–7]. Iron and aluminium oxyhydroxides cement these ancient surfaces, which have persisted under prolonged tropical weathering and form isolated plateaux with shallow upland lakes at approximately 695–765 m above sea level [4–9]. The lake basins are largely hydrologically disconnected; their water levels therefore respond strongly to the balance between precipitation and evaporation, with rainfall concentrated from November to May and a pronounced dry period from June to October [8–10]. This setting combines geological stability, seasonal hydrological forcing and strong mineral control within a compact group of tropical aquatic ecosystems. The canga plateaux support montane savanna vegetation embedded within the surrounding Amazon rainforest [1,4,15], and the Carajás mineral province contains one of the world's largest high-grade iron ore reserves [1,3].

Carajás canga-lake sediments comprise compact mixtures of organic and inorganic material enriched in iron, aluminium, phosphorus and trace elements, including selenium and vanadium [2,11–14]. Detrital supply, water-level variation, organic-matter input and authigenic mineral formation influence the distribution and reactivity of these constituents [2,11,12]. Because iron oxyhydroxides are both redox-active and strongly sorptive, seasonal changes in oxygen penetration and mineral reactivity can alter the retention and release of phosphorus, organic compounds and trace elements. Nutrient availability, pH, dissolved oxygen and chlorophyll-a also vary seasonally [8,14]. The lakes additionally support endemic and spatially restricted biota, including the quillworts *Isoetes cangae* and *Isoetes serracarajensis* [15,16], emphasizing their conservation and ecological significance. Within this restricted flora, *I. cangae* is confined to Lake Amendoim, whereas *I. serracarajensis* has a broader distribution that includes Três Irmãs and Violão [16]. This species-level contrast provides a conservation-relevant basis for future tests of whether sediment microbiomes and biogeochemical conditions contribute to aquatic-plant habitat specificity.

Ferruginous lake sediments are dynamic redox mosaics rather than passive mineral deposits. Microbial oxidation and reduction reactions can couple organic-matter mineralization to carbon, methane, nitrogen, sulfur and iron transformations, while repeated flooding and sediment exposure reorganize oxygen penetration, electron-acceptor availability and mineral surfaces [8,14,17]. Shotgun metagenomics can resolve community composition together with encoded metabolic potential, and genome-resolved reconstruction can associate pathways with uncultivated populations. Nevertheless, genes and reconstructed modules indicate potential rather than measured activity and must be interpreted with hydrological and geochemical context. Despite extensive geological, sedimentological and limnological study of Carajás, the taxonomic and functional diversity of lateritic-lake sediment microbiomes, their genome-resolved representatives and their seasonal relationship to iron-rich geochemistry remain insufficiently resolved. Seasonal shifts in nutrient profiles, catchment conditions and phytoplankton biomass further illustrate that the biological response of these lakes is coupled to hydrology and limnological variability [8,14,15].

Here, we used shotgun metagenomics to characterize sediments from Amendoim (AM), Violão (VI), Três Irmãs (TI) and Três Irmãs Adjacent (TIA) during dry and rainy periods. We integrated coding-sequence (CDS) taxonomic profiling, alpha- and beta-diversity analyses, non-metric multidimensional scaling (NMDS), differential abundance, Kyoto Encyclopedia of Genes and Genomes (KEGG) module reconstruction, curated KEGG Orthology (KO) biomarkers, FeGenie-supported iron-gene annotation, redundancy analysis (RDA) of physicochemical covariates, metagenome-assembled genome (MAG) reconstruction and descriptive directional contrasts with external iron-rich records. The resulting atlas was designed to establish a reproducible baseline for evaluating how hydrological seasonality and ferruginous sediment chemistry are associated with microbial community structure and biogeochemical potential in Amazonian canga lakes.

Materials and Methods

Study area and sampling design

Violão, Amendoim and Três Irmãs lakes are perennial highland systems located in the Carajás National Forest, southeastern Amazon, Brazil, on ferruginous lateritic plateaus at approximately 695–765 m above sea level [8,9,14,15] (Figure 1). Três Irmãs is the largest system and becomes connected during the rainy season; during the dry season, an adjacent water body is considered here as Três Irmãs Adjacent Lake. Sediments were collected during the rainy season (March) and the dry season (September). Sampling points were selected from longitudinal profiles informed by bathymetric maps and drainage patterns. Two sites were sampled in AM, VI and TIA, and four sites were sampled in TI because of its larger area (Figure 1; Supplementary Figure 30). Surface sediments (30–40 mL) were collected with an Ekman–Birge dredge, transferred to sterile 50 mL polypropylene tubes and stored at −20 °C until DNA extraction. Available physicochemical information for these lakes was integrated from previous limnological and geochemical surveys [14]. Exact geographic coordinates for all sampling positions are provided in the “Sampling coordinates” sheet of Supplementary Table 1; Figure 1 and Supplementary Figure 30 show their spatial distribution.

DNA extraction, library preparation and shotgun sequencing

Total DNA was extracted from 20 sediment samples, each containing approximately 250 mg of sediment, using the PowerSoil DNA Isolation Kit (MoBio Laboratories, USA) according to the manufacturer’s instructions. Duplicate extractions were performed for each sample. DNA quality and concentration were assessed by 1% agarose gel electrophoresis and Qubit 3.0 fluorometry (Life Technologies, Thermo Fisher Scientific). Shotgun paired-end metagenome libraries were prepared from approximately 50 ng DNA using enzymatic random fragmentation and the SureSelectQXT kit (Agilent Technologies). Fragmented DNA was purified with AMPure XP beads, amplified with Illumina-adapter primers, purified again, quantified with Qubit and evaluated using an Agilent 2100 Bioanalyzer with the High Sensitivity DNA kit. Libraries were diluted, pooled and sequenced on an Illumina NextSeq 500 platform using a NextSeq 500 v2 high-output 300-cycle kit.

The metagenomic sequencing project was registered in the Genomes OnLine Database (GOLD) under project application number Gp0646171. The raw metagenomic sequencing data are publicly available in the European Nucleotide Archive under study accession ERP137391 and run accessions ERR9980190–ERR9980209.

Read processing, assembly and functional annotation

Raw reads were quality-filtered with PRINSEQ, retaining reads with an average quality score ≥20, and quality summaries were generated with FastQC [18,19]. Trimmed reads were assembled de novo with SPAdes v3.7.0 in metagenomic mode using the parameters --meta --only-assembler -k 21,33,55,77,121 [20]. Assembly quality was evaluated with MetaQUAST/QUAST v3.0, including N50, L50 and largest-contig metrics [21]. Predicted coding sequences and functional annotation of scaffolds were generated with the IMG/MER pipeline [22]. KEGG modules were reconstructed using a local KAAS-based workflow, and module completeness was evaluated with an adapted KEMET procedure [23–26]. Complete, one-block-missing, two-block-missing and incomplete modules were summarised across metagenomes, and final module-completeness matrices were retained as source data (Supplementary Figure 38; Supplementary Tables 3 and 6).

Iron-metabolism potential was evaluated with FeGenie, which annotates genes associated with iron acquisition, storage, oxidation, reduction and iron-gene neighbourhoods in genomes and metagenomes [27]. FeGenie gene categories were summarised at both metagenome and MAG levels (Supplementary Figures 15 and 16).

Taxonomic profiling and community diversity

IMG/MER-predicted coding sequences were assigned taxonomically with Kaiju v1.9.0 using the nr_euk database, which includes proteins from Archaea, Bacteria, viruses, fungi and microbial eukaryotes [28]. Kaiju outputs were converted into OTU and taxonomy tables with the kaiju2-data-parser workflow, and tables were processed in R with dplyr and phyloseq [29,30]. For alpha-diversity and rarefaction, the CDS OTU matrix was rarefied to a fixed retained depth of 32,999 CDS, corresponding to the smallest predicted-CDS sample depth and allowing all 20 samples to be retained. The same fixed-depth rarefaction was used for rarefaction curves and for Observed OTUs, Chao1 richness and Shannon diversity metrics. Bar plots, box plots and statistical tests were generated with ggplot2 [31].

Beta diversity was evaluated from sample-wise relative taxonomic proportions transformed by square root, followed by Bray–Curtis dissimilarity. Two-dimensional non-metric multidimensional scaling (NMDS) was fitted with 20 random initializations, a maximum of 1,000 iterations, normalized Stress-1 and random seed 42. All 20 dry- and rainy-season metagenomes were retained. Lake, season and lake–season effects were audited with 999-permutation PERMANOVA, together with permutation tests of multivariate dispersion. Genus-level community profiles were integrated with physicochemical measurements available for ten sampling positions. Because chemistry was measured independently at position rather than season level, dry and rainy taxonomic counts were pooled by position before Hellinger transformation; thus each RDA point represents one position and accounts for its two seasonal metagenomes. Environmental predictors were standardized, and LOI, SiO_2, Al_2O_3, total sulfur, Cu and Pb were retained. Collinearity was reported by variance-inflation factors. Global and axis-specific RDA significance used 999 permutations with seed 42. Owing to the small residual degrees of freedom and non-significant global tests, the RDA was interpreted as exploratory (Figures 4D and 5D; Supplementary Figure 17).

Curated KO biomarkers for biogeochemical cycling and iron metabolism

To complement KEGG module completeness, we developed a marker-based framework focused on key KEGG Orthology (KO) biomarkers for carbon fixation, methane metabolism, nitrogen cycling, sulfur cycling, oxygenic and anoxygenic photosynthesis, anaerobic oxidative phosphorylation and iron metabolism. We started from a 77-marker list for carbon fixation, nitrogen and sulfur cycling [33] and expanded it to 195 KO biomarkers representing seven evaluated carbon-fixation pathway categories, methane metabolism, nitrogen cycling, photosynthesis, anaerobic ammonium oxidation, DMSO-linked anaerobic oxidative phosphorylation and alternative nitrogenase forms. Markers also included carbon monoxide dehydrogenase, nitrite reductase, nitric oxide reductase, nitrate reductase and phosphate acetyltransferase. Of these 195 marker KOs, 171 were detected in the IMG/MER functional annotation dataset for AM, TI and VI sediments. These biomarkers indicate the genetic potential for a wide range of transformation processes in the carbon (C), methane (CH_4), nitrogen (N) and sulfur (S) cycles, as well as for oxygenic and anoxygenic photosynthesis and anaerobic oxidative phosphorylation using dimethyl sulfoxide (DMSO) as a terminal electron acceptor. In terms of carbon cycling, all seven known carbon fixation pathways were present in each lake, regardless of season (Figure 8; Supplementary Figures 6, 8, 9, 32 and 33; Supplementary Table 8). The marker panel was treated as a complementary line of evidence rather than as a replacement for module reconstruction. The KO-based strategy also enabled consistent mining of orthologous functions from existing IMG/MER and KAAS annotations without requiring a separate local alignment or profile-HMM search for every marker at the downstream screening stage. The iron-marker set included 132 KOs associated with iron metabolism and redox cycling. These markers were generated from FeGenie category keywords and curated literature-supported KO assignments and were grouped into 12 categories: heme synthesis, heme transport, iron acquisition, iron regulation, iron oxidation, iron reduction, iron storage, iron transport, magnetosome formation, siderophore secretion, siderophore synthesis and siderophore transport (Supplementary Figure 69; Supplementary Figures 7, 10, 11, 15, 16 and 36; Supplementary Tables 4 and 5).

Within-Amazonian differential abundance and descriptive cross-study mean-ratio analysis

Within the Amazonian dataset, differential abundance of KO biomarkers was evaluated across prespecified pairwise lake-season comparisons using DESeq2 and ALDEx2. KO counts were normalized using the geometric mean of pairwise ratios (GMPR), and DESeq2 size factors were estimated before fitting negative-binomial models with Wald tests and parametric dispersion estimation [34–36]. Unless explicitly reported as false-discovery-rate-adjusted q values, screening summaries were based on nominal $P < 0.05$ together with absolute effect-size thresholds. The sign of each DESeq2 $\log_2$ fold change indicated which member of the corresponding lake-season comparison had the higher estimated KO abundance.

The within-Amazonian analysis comprised 12 prespecified pairwise lake-season comparisons and 2,052 KO-level statistical tests. Only two tests remained significant at $q < 0.05$, both involving the mercury reductase marker K00320 (mer). No test within the iron-associated KO subset reached $q < 0.05$. Consequently, Figure 8 presents the KOs with the largest positive or negative $\log_2$ fold changes as descriptive summaries of the strongest within-Amazonian differences. These results should not be interpreted as evidence of widespread FDR-significant changes in biogeochemical pathways.

To compare the Amazonian dataset with external iron-rich datasets, we performed a separate descriptive cross-study mean-ratio analysis, referred to here as a directional contrast analysis. For each KO, this analysis determined which of the two groups had the higher mean relative representation and the magnitude of that difference on a $\log_2$ scale. In other words, the analysis was designed to answer whether a given KO was proportionally more represented, on average, in the Amazonian lateritic-lake metagenomes or in the external iron-rich records.

The comparison included the 20 Amazonian lateritic-lake metagenomes and 67 external iron-rich metagenomic and metatranscriptomic records obtained from IMG/JGI. The corresponding project identifiers, environmental classifications, omics layers, study groupings and references are provided in Supplementary Table 8. Because these external records originated from different studies, environments, sequencing designs and omics layers, the analysis was used only as a descriptive cross-study comparison. It was not an inferential statistical test and did not generate P values, confidence intervals or FDR-adjusted q values.

For each sample, the count of each KO was divided by the total count of all KOs included in the corresponding marker panel, producing a within-sample marker-panel relative abundance. The mean relative abundance of each KO was then calculated separately for the Amazonian and external groups. The directional contrast for each KO was calculated as:

$$\text{Directional contrast} = \log_2\left(\frac{\text{mean relative abundance in Amazonian lakes} + \varepsilon}{\text{mean relative abundance in external records} + \varepsilon}\right)$$

where ε represents a small pseudocount added to avoid undefined ratios when a KO had a mean relative abundance of zero in one of the groups. A positive contrast indicated that the KO had a higher mean relative representation in the Amazonian lateritic-lake metagenomes, whereas a negative contrast indicated higher mean relative representation in the external iron-rich records. For example, a value of +1 indicated an approximately twofold higher mean relative representation in the Amazonian group, whereas a value of −1 indicated an approximately twofold higher mean relative representation in the external group. The absolute value represented the magnitude of the descriptive difference between the two group means.

These directional contrast values are therefore descriptive $\log_2$ ratios of group means and should not be confused with the model-based $\log_2$ fold changes estimated by DESeq2 for the within-Amazonian comparisons. For visualization, Supplementary Figures 68 and 69 show the 12 KOs with the strongest positive contrasts toward the Amazonian group and the 12 KOs with the strongest negative contrasts toward the external group after prevalence and abundance filtering. Complete KO-level contrast tables and broader descriptive summaries are provided in Supplementary Figures 35 and 36 and Supplementary Table 8.

Generation of main and supplementary figures

All main and supplementary figures were generated from the processed matrices and supplementary tables described above using Python 3.10 with pandas, NumPy, SciPy, scikit-learn and Matplotlib; Plotly was used only for the corresponding interactive application views. Unless stated otherwise, categorical summaries were calculated from exact counts or relative-abundance matrices, and stochastic procedures used a fixed random seed of 42. Static figures were exported at a minimum of 300 dpi in PNG format and as vector PDF and SVG files. Figure generation did not alter the underlying values: the plotted matrices were also retained as tabular source data. Figures sharing the same biological categories used fixed colour assignments so that a taxon, environmental group or module-completeness state had the same colour across the manuscript, supplementary material and interactive application.

Sequencing, diversity, ordination, MAG and environmental summaries

Supplementary Figure 1 was generated from the sequencing, assembly, gene-prediction and taxonomic-classification statistics reported for each metagenome. Numeric variables were grouped by lake-season combination and displayed as boxplots with the corresponding sample values. For Supplementary Figures 2 and 4, the CDS abundance matrix was rarefied to 32,999 CDS per sample, the minimum retained sample depth. Rarefaction used deterministic multivariate-hypergeometric subsampling with seed 42, and curves were evaluated across increasing depths up to 32,999 CDS. Observed richness was the number of taxa with non-zero counts after subsampling; Chao1 was calculated from observed richness and the numbers of singleton and doubleton taxa; and Shannon diversity was calculated as $-\Sigma[p_i \ln(p_i)]$, where p_i is the rarefied relative abundance of taxon i. Supplementary Figure 2 shows the resulting rarefaction curves, whereas Supplementary Figure 4 summarizes the final Observed, Chao1 and Shannon values by lake-season group using boxplots and individual sample points. Supplementary Figure 3 was generated from Bray-Curtis dissimilarities calculated on square-root-transformed sample-wise relative abundances derived from the taxonomic matrix; two-dimensional non-metric multidimensional scaling used 20 initial configurations, a maximum of 1,000 iterations and seed 42. Supplementary Figure 5 combined the MAG abundance matrix with completeness and contamination values from Supplementary Table 7; bubble area represented relative abundance, and the accompanying quality displays summarized completeness and contamination. Supplementary Figure 17 combined the genus-level RDA scores and vectors described above with a physicochemical heatmap. Environmental variables in the heatmap were standardized independently as z scores, $z_{ij} = (x_{ij} - mean_j)/SD_j$, to compare relative deviations among sampling positions despite differences in measurement units.

KO-marker matrices, heatmaps, z scores and summary plots

KO figures were generated separately for the 195-member biogeochemical marker panel and the 132-member iron-associated marker panel. Exact KO counts were retained for the raw-count visualizations. Supplementary Figure 32 displays the untransformed biogeochemical-marker count matrix. For row-standardized KO heatmaps, each KO was standardized across samples as $z_{ij} = (x_{ij} - mean_i)/SD_i$, where x_{ij} is the value for KO i in sample j, $mean_i$ is the mean of that KO across the displayed samples and SD_i is its population standard deviation; rows with zero variance were assigned a z score of zero. This transformation emphasizes the relative distribution of each marker across samples and does not compare absolute abundance among different KOs. Supplementary Figure 33 displays the resulting row-z-score matrix for the Amazonian biogeochemical markers. The same row-standardization procedure was used for the common-taxon heatmap in Supplementary Figure 31 and for any application view explicitly labelled as a row-z-score heatmap; the exact non-standardized values were retained in the associated source tables.

For abundance-ranking summaries, mean counts were calculated across the records represented in each panel and markers were ordered from highest to lowest mean abundance (Supplementary Figures 8 and 11). Category-level distributions were generated by assigning each marker to its curated metabolic or iron-related category and displaying log1p-transformed normalized counts as boxplots, with individual markers forming the observations within each category (Supplementary Figures 9 and 10). Supplementary Figures 6 and 7 display selected bidirectional descriptive contrasts for the biogeochemical and iron-marker panels, respectively, and Supplementary Figure 12 combines the strongest contrasts from both marker sets. Supplementary Figures 15 and 16 were generated from FeGenie category counts: the heatmap displays log1p-transformed counts for each metagenome, and the boxplots summarize the same category counts by lake with individual sample values. Supplementary Figures 34-36 were generated from the curated external-record metadata and marker matrices. Supplementary Figure 34 counts records by environmental group and omics layer. Supplementary Figures 35 and 36 show the complete descriptive KO mean-ratio landscapes for the biogeochemical and iron panels, respectively, using the directional log2 mean ratios defined above; bar direction identifies the group with the higher mean relative representation, and no inferential P value is associated with these panels. Supplementary Figures 68 and 69 display the 12 strongest positive and 12 strongest negative directional contrasts after the prevalence and abundance filters described above.

Taxonomic profiles, heatmaps and cross-study overlap figures

For the Amazonian CDS-derived taxonomic figures, OTU counts were joined to the taxonomy table, separated into Bacteria and Archaea and aggregated independently at phylum, class, order, family, genus and species levels. Within each domain and sample, aggregated counts were divided by the total classified count for that domain and multiplied by 100. Main Figures 2-5 and Supplementary Figures 43-66 were produced from these domain-specific relative-abundance matrices. For each 100% stacked barplot, taxa were ranked by their total abundance across samples, the 24 most abundant taxa were retained and all remaining taxa were summed as Others; each sample was then renormalized to 100%. For the paired heatmaps, the 30 taxa with the highest total abundance were retained and displayed as log10(relative abundance [%] + 1). Low values were mapped to red, intermediate values to pale colours and high values to blue using a reversed coolwarm scale. The exact untransformed percentages were retained for every heatmap.

Supplementary Figures 14 and 18-25 were generated from the curated GTDB group-by-taxon tables for the 67 external iron-rich records. Counts were aggregated by environmental group and omics layer, converted to within-group relative abundance, ranked at the displayed taxonomic level and represented as 100% stacked profiles with residual taxa combined as Other taxa. Supplementary Figures 26-28 were set-overlap summaries. Presence was defined by a non-zero abundance in the corresponding combined-assembly, metagenomic or metatranscriptomic layer; intersections and unions were calculated from the resulting taxon sets at phylum, order and family levels. Supplementary Figure 29 is a schematic representation of the computational workflow and data flow. Supplementary Figure 30 plots the verified latitude and longitude coordinates from the “Sampling coordinates” sheet of Supplementary Table 1. Supplementary Figure 31 was generated from taxa present across the selected iron-rich environmental groups; the group-level abundance matrix was row-standardized using the z-score calculation defined above.

KEGG/KEMET module-completeness matrices and figures

KEMET reportKMC files were parsed as headerless tables containing module identifier, module name, categorical status, completed-block information, missing or alternative KOs and detected KOs. MAG identifiers and IMG/JGI metagenome identifiers were normalized before matrices were assembled. For each biological unit, a module was classified as Complete when every required block was detected, 1 block missing when exactly one required block was unresolved, and Incomplete / >=2 blocks missing for all remaining incomplete states. These three states were encoded as 2, 1 and 0, respectively, solely for plotting and were displayed using fixed colours: Complete, green (#2E7D32); 1 block missing, blue (#1565C0); and Incomplete / >=2 blocks missing, red (#C62828). No clustering, interpolation or imputation was applied, and all module rows and all samples/MAGs were retained.

Supplementary Figure 37 was generated from the MAG-by-module status matrix, with columns labelled by species assignment and MAG number. Supplementary Figure 38 used the corresponding matrix for all 20 Amazonian metagenomes. For the 67 external IMG/JGI records, evaluated KEGG modules were reconstructed from the curated KO presence matrix using the same required-block logic, producing the matrix shown in Supplementary Figure 40. Supplementary Figure 41 reports, for each external record, the number of modules assigned to each of the three states. Supplementary Figure 67 was produced by concatenating the Amazonian and external status matrices by matched KEGG module identifier; it provides a cross-environment view while preserving the original status assigned in each source matrix. Supplementary Figure 42 was generated by grouping KO counts by lake-season combination and plotting the mean abundance of the selected markers for each group. In every module-completeness display, modules were retained only when at least one sample, MAG or external record was classified as Complete; modules incomplete in every column were excluded.

Interactive platform implementation and reproducible data delivery

A local Streamlit application was developed in Python to provide readers with an interactive interface for visualizing and exploring the processed results generated by the analytical workflows described above. The application was organized into modular components for loading supplementary tables, taxonomic and functional profiles, KEGG/KEMET module-completeness matrices, KO-marker panels, physicochemical and ordination results, MAG quality and abundance summaries, FeGenie outputs and antiSMASH biosynthetic-gene-cluster results. A centralized data-loading layer reads the packaged CSV, TSV and XLSX source tables, validates file availability and required columns, and passes the processed matrices to Streamlit tables and Plotly visualizations. The interface provides searching, filtering, interactive inspection and downloading of the same processed values used to generate the static manuscript and supplementary figures.

The application does not replace or alter the analytical procedures used to produce the article results. It reads the synchronized processed matrices generated by the standalone scripts, including the KEGG/KEMET status and completeness-score matrices, domain-separated taxonomic relative-abundance tables, KO count and standardized matrices, FeGenie summaries, MAG metadata and antiSMASH region tables. It can therefore be executed locally by readers as an optional visualization and data-exploration resource, whereas all canonical static figures remain independently reproducible from the packaged inputs and figure-generation scripts. The source code, environment files, execution instructions and processed data structure are available in the project repository at https://github.com/mattoslmp/GangaMetaG-IronMetagenomicAtlas.

MAG reconstruction, quality assessment and annotation

MAGs were reconstructed using a co-assembly strategy in which reads from all samples were pooled before assembly. Co-assembly was performed with metaSPAdes v3.15.1 using -k auto, co-correction and default parameters. Genome binning used the MetaWRAP v1.3 binning module integrating metaBAT2, CONCOCT and MaxBin2 [37]. Bins were consolidated into a refined non-redundant MAG set using bin-refinement with Binning_refiner and DAS_Tool, requiring minimum completeness of 50% and maximum contamination of 5%. Reassembly was performed with the MetaWRAP Reassemble_bins module.

MAG quality was assessed with CheckM v1.2.1 and BV-BRC based on completeness and contamination [38,39]. Taxonomic annotation used GTDB-Tk v2.1.1 with GTDB R07-RS207, CheckM, MetaWRAP Classify_bins, BV-BRC, Kaiju and rRNA sequences extracted from BV-BRC bin-contig annotations [28,37–40]. MAG quality was interpreted following MIMAG criteria: high quality, completeness ≥90% and contamination ≤5%; good or intermediate quality, completeness ≥70% and contamination ≤10%; and medium quality, completeness ≥50% and contamination ≤10% [39,41]. Species-level assignments were treated conservatively in light of ANI-based species-boundary concepts [42]. The biosynthetic gene cluster (BGC) potential of the recovered metagenome-assembled genomes (MAGs) was evaluated using the supplied antiSMASH output directories, including the original `index.html` reports and JSON files. In the final MAG/bin dataset, bin identifiers were used as MAG identifiers. The antiSMASH `index.html` reports were parsed to extract the numbers of BGC regions and protoclusters, predicted products, biosynthetic categories, and MIBiG matches, when available. The parser retained all MAG/bin identifiers from MAG.1 to MAG.50. MAG.2, MAG.5, MAG.20, MAG.32, MAG.44, MAG.47, and MAG.49 were recorded as having no BGC regions detected in the final antiSMASH dataset provided. The antiSMASH outputs were generated using antiSMASH v8.0.4 and are summarized in Supplementary Table 11 [43].

All MAGs were annotated using the Bacterial and Viral Bioinformatics Resource Center (BV-BRC) [38] and are publicly available in the BV-BRC workspace at https://www.bv-brc.org/workspace/mattoslmp@patricbrc.org/Lakes-Canga/metagenomas.

ENA sample aliases and analysis accessions for MAG.1–MAG.50 are provided in Supplementary Table 7.

Results

Sequencing of 20 sediment metagenomes from AM, VI, TI and TIA generated approximately 66 million high-quality paired-end reads with an average read length of 150 bp (Supplementary Table 1). Assembly and gene prediction produced sample-specific CDS datasets, including 32,999–420,093 predicted CDS per sample, and comparative sequencing, assembly and annotation metrics were summarised across lake-season groups (Supplementary Figure 1; Supplementary Table 1). Kaiju classification of IMG/MER predicted CDSs detected 25,534 bacterial and 1,232 archaeal assignments across the dataset (Supplementary Table 1). Bacteria and Archaea dominated the classified fraction, whereas viruses and eukaryotes together represented less than 1% of assignments. A large fraction of sequences remained unclassified, ranging approximately from 56.2% to 60.4%, indicating that these sediments contain substantial underexplored dark microbial diversity.

Phylum-level taxonomic profiles showed consistent representation of Proteobacteria, Chloroflexi, Acidobacteria, Euryarchaeota, Nitrospirae, Candidatus Aminicenantes, Planctomycetes, Firmicutes, Actinobacteria, Verrucomicrobia, Bacteroidetes, Candidatus Thermoplasmatota and Candidatus Rokubacteria across lake-season combinations (Figures 2 and 3). Bacterial genus-level profiles highlighted Anaeromyxobacter as a prominent bacterial genus and included Bradyrhizobium, Candidatus Sulfotelmatobacter, Candidatus Sulfopaludibacter, Clostridium, Geobacter, Mycobacterium, Nitrospira, Paenibacillus, Pseudolabrys, Ralstonia, Streptomyces, Syntrophorhabdus and Syntrophus, whereas archaeal genus-level profiles included Candidatus Bathyarchaeota, Candidatus Nitrosotalea, Candidatus Methanoperedens, Methanoregula, Methanosaeta, Methanolinea, Methanothrix and Nitrosopumilus-related lineages (Figures 4 and 5). Cross-study taxonomic context and overlap analyses are provided in Supplementary Figures 14, 18-31. Differential abundance analyses showed that taxonomic differences were more pronounced between lakes than between the dry and rainy seasons within the same lake. Several enriched taxa in dry-season comparisons belonged to Archaea or anaerobic sediment-associated Bacteria, including Candidatus Nitrosotalea, Candidatus Micrarchaeum acidiphilum, Candidatus Methanoperedens nitroreducens, Methanoregulaceae archaeon, Methanoregula sp., Candidatus Bilamarchaeum dharawalense and multiple Anaeromyxobacter representatives (Figure 6; Supplementary Figure 39). Rainy-season contrasts included Methanophagales archaeon ANME-1-THS, Nitrosopumilales archaeon, Candidatus Nitrososphaera gargensis, Methanosaeta sp. PtaU1.Bin060 and Methanoregulaceae-related taxa. Rarefaction curves supported adequate sampling depth for the detected CDS-based diversity and were regenerated with all 20 samples retained to a fixed rarefaction depth of 32,999 CDS (Supplementary Figure 2). At this retained depth, Observed OTUs ranged from 1,773 to 3,503, Chao1 richness from 4,687.4 to 8,104.5 and Shannon diversity from 2.62 to 3.63 across lake-season samples. The recalculated alpha-diversity panel showed comparatively higher richness in several AM rainy and TI dry or rainy samples and lower richness in VI dry samples, while TIA samples remained intermediate and relatively compact across metrics (Supplementary Figure 4). Bray-Curtis NMDS based on CDS-derived profiles did not show strict lake-wise clustering; grouped and individual-sample ordinations instead supported partial overlap and lake-season heterogeneity (Figures 4C and 5C; Supplementary Figure 3). The domain-specific global RDA models were not significant (Bacteria: constrained R^2 = 0.671, adjusted R^2 = 0.012, pseudo-F = 1.02, permutation P = 0.537; Archaea: constrained R^2 = 0.606, adjusted R^2 = −0.183, pseudo-F = 0.77, permutation P = 0.698). Relationships of genus-level variation with LOI, SiO_2, Al_2O_3, total sulfur, Cu and Pb are therefore presented as exploratory covariation (Figures 4D and 5D; Supplementary Figure 17). Functional reconstruction detected broad metabolic potential across the sediments. Complete and near-complete KEGG modules included the reductive citric acid cycle, dicarboxylate/4-hydroxybutyrate cycle, 3-hydroxypropionate/4-hydroxybutyrate cycle, 3-hydroxypropionate cycle, Wood-Ljungdahl pathway, acetate production via acetyl-CoA, the Calvin-Benson-Bassham cycle, formaldehyde assimilation, nitrogen fixation, assimilatory and dissimilatory nitrate reduction, assimilatory sulfate reduction, cytochrome bc1 complex, cytochrome c oxidase and heme/siroheme biosynthesis modules. MAG-level and metagenome-level KEGG module completeness patterns are summarised in Supplementary Figures 37 and 38 and Supplementary Tables 3, 6 and 9.

The curated KO-marker panel detected 171 of the 195 biogeochemical biomarkers in Amazonian sediments, indicating genetic potential for carbon fixation, methane metabolism, nitrogen cycling, sulfur cycling, photosynthesis, and anaerobic oxidative phosphorylation. Within-Amazonian KO contrasts revealed large-magnitude LFCs for markers associated with sulfur, nitrogen, carbon-fixation, and respiratory pathways. These included *sir*, associated with sulfite reduction in assimilatory sulfur metabolism; *nrfH*, involved in respiratory nitrite reduction to ammonium through dissimilatory nitrate reduction to ammonium (DNRA); *cysJ* and *cysI*, associated with NADPH-dependent assimilatory sulfite reduction; *pccA*, involved in propionyl-CoA carboxylation associated with propanoate metabolism and the 3-hydroxypropionate carbon-fixation cycle; *dmsC*, associated with dimethyl sulfoxide reduction during anaerobic respiration; *napA*, involved in the periplasmic reduction of nitrate to nitrite; *norC*, associated with nitric oxide reduction during denitrification; *pta*, involved in the conversion of acetyl-CoA to acetate through the phosphate acetyltransferase–acetate kinase pathway; *nosZ*, associated with the reduction of nitrous oxide to dinitrogen during the final step of denitrification; *soxY*, involved in thiosulfate oxidation through the Sox sulfur-oxidation pathway; and *nirS*, associated with nitrite reduction to nitric oxide during denitrification. These contrasts were observed across dry- and rainy-season comparisons (Figure 8; Supplementary Figures 6, 8, 9, 32, and 33). Complete descriptive KO-contrast landscapes and cross-study marker comparisons are provided in Supplementary Figures 12, 34, and 35. Expanded marker summaries are presented in Supplementary Figures 6 and 7. Iron-related analyses identified 132 iron-associated KOs and FeGenie-supported iron-related gene categories linked to iron acquisition, siderophore metabolism, heme transport and synthesis, iron storage, iron oxidation, and iron reduction. Iron-marker summaries across Amazonian and external iron-rich records are shown in Supplementary Figures 7, 10, 11, 15, 16, and 36. At the MAG level, FeGenie detected iron-related genes in *Anaeromyxobacter* sp. Red267, a Desulfobacterales/Deltaproteobacteria bacterium, an Acidobacteria bacterium, a Burkholderiales bacterium, and *Nitrospirales bacterium* LBB_01, thereby linking community-wide iron-marker signals to genome-resolved candidates.

Descriptive cross-study comparison showed how marker profiles differed between Amazonian lateritic lakes and external iron-rich records. Supplementary Figure 68 summarises the 12 biogeochemical KOs with the strongest contrast toward the Amazonian lakes and the 12 with the strongest contrast toward the external record. KOs with higher mean representation in Amazonian canga lakes included *fdhA* (coenzyme F420-dependent formate dehydrogenase alpha subunit; formate oxidation and CO_2 reduction in methane metabolism), *VnfE* (nitrogenase vanadium-cofactor synthesis protein VnfE; vanadium-dependent nitrogen fixation), *hdrA1* (heterodisulfide reductase subunit A1; electron-bifurcating energy conservation in methanogenesis), *hdrB1* (heterodisulfide reductase subunit B1; heterodisulfide reduction and energy conservation in methanogenesis), *hdrC1* (heterodisulfide reductase subunit C1; heterodisulfide reduction and energy conservation in methanogenesis), *norV* (anaerobic nitric oxide reductase flavorubredoxin; nitric oxide reduction during anaerobic respiration and nitrosative-stress detoxification), *dmsD* (putative dimethyl sulfoxide reductase chaperone;

maturation of the DMSO reductase complex during anaerobic respiration), *fdhB* (coenzyme F420-dependent formate dehydrogenase beta subunit; formate oxidation and CO_2 reduction in methane metabolism), *comC* (L-2-hydroxycarboxylate dehydrogenase [NAD+]; coenzyme M biosynthesis associated with methanogenesis), *mcrA* (methyl-coenzyme M reductase alpha subunit; terminal methane formation during methanogenesis and reverse methanogenesis), and *mtrF* (tetrahydromethanopterin S-methyltransferase subunit F; methyl-group transfer and energy conservation during methanogenesis). Collectively, these markers spanned methanogenesis, nitrogen fixation, denitrification, dimethyl sulfoxide-associated respiration, and the Wood–Ljungdahl pathway (Supplementary Figures 6, 12, 35, and 68).

KOs with higher mean representation in external iron-rich records included *psaB* (photosystem I P700 chlorophyll *a* apoprotein A2; oxygenic photosynthesis through photosystem I), *mdh1* (methanol dehydrogenase [cytochrome *c*] subunit 1; methanol oxidation and methylotrophic carbon metabolism), *mdh2* (methanol dehydrogenase [cytochrome *c*] subunit 2; methanol oxidation and methylotrophic carbon metabolism), *smtA1* (succinyl-CoA:(S)-malate CoA-transferase subunit A; autotrophic carbon fixation through the 3-hydroxypropionate bicycle), *smtB* (succinyl-CoA:(S)-malate CoA-transferase subunit B; autotrophic carbon fixation through the 3-hydroxypropionate bicycle), *pmoA-amoA* (methane/ammonia monooxygenase subunit A; methane oxidation or ammonia oxidation during nitrification), *pmoB-amoB* (methane/ammonia monooxygenase subunit B; methane oxidation or ammonia oxidation during nitrification), *pmoC-amoC* (methane/ammonia monooxygenase subunit C; methane oxidation or ammonia oxidation during nitrification), *pufL* (photosynthetic reaction-centre L subunit; anoxygenic photosynthesis in purple bacteria), and *pucA* (light-harvesting protein B-800–850 alpha chain; light harvesting during anoxygenic photosynthesis) (Supplementary Figure 68).

For iron-associated KOs, Supplementary Figure 69 summarises the 12 markers with the largest absolute directional contrasts in each comparison. Amazonian lateritic lakes showed stronger directional signals for *qrcB* (menaquinone reductase molybdopterin-binding-like subunit; respiratory electron transfer associated with microbial iron reduction), *qrcC* (menaquinone reductase iron–sulfur-cluster-binding subunit; respiratory electron transfer associated with microbial iron reduction), *qrcD* (menaquinone reductase integral membrane subunit; membrane-associated respiratory electron transfer linked to iron reduction), *fnr* (ferredoxin/flavodoxin–NADP+ reductase; redox electron transfer associated with iron-reduction metabolism), heme oxygenase TB11.2 (MhuD-type, mycobilin-producing heme oxygenase; heme degradation and iron acquisition), putative heme transporter K20469 (putative heme-transport protein; heme uptake and iron acquisition), *soxC* (sulfane dehydrogenase subunit SoxC; sulfur oxidation through the Sox pathway), *bfrB* (bacterioferritin B; intracellular iron oxidation, mineralisation, and storage), *mmpL3* (trehalose monomycolate/heme transporter; heme transport, iron acquisition, and mycobacterial cell-envelope biogenesis), *ahbC* (Fe-coproporphyrin III synthase; alternative siroheme-dependent heme-biosynthesis pathway), *ahbD* (AdoMet-dependent heme synthase; terminal step of the alternative siroheme-dependent heme-biosynthesis pathway), and *Cyc2* (iron:rusticyanin reductase; ferrous-iron oxidation and electron transfer) (Supplementary Figures 7, 11, 14, 36, and 69).

External iron-rich records showed stronger contrasts for heme uptake and heme oxygenase markers, siderophore synthesis, iron-sulfur cluster assembly, cytochrome c oxidase, iron transport, haemoglobin glbN and the catecholate siderophore receptor fiu (Supplementary Figure 69). Metadata availability, study grouping, verified coordinates and the computational workflow are documented in Supplementary Figures 13, 29, 30 and 34.

Genome-resolved analysis recovered 50 non-redundant MAGs that met medium- to high-quality criteria. Completeness ranged from approximately 50.83% to 100% and contamination from 0% to 4.9%. Binning results included outputs from metaBAT2, CONCOCT and MaxBin2 followed by refinement and reassembly. Identified genomic bins were distributed mainly across Crenarchaeota, Chloroflexota, Acidobacteriota, Thermoplasmatota and Asgardarchaeota, with additional representatives from Cyanobacteria, Nitrospirota, Myxococcota, Proteobacteria, Desulfobacterota, Actinobacteriota, Bacteroidota and other phyla (Figure 7; Supplementary Table 7). The corresponding ENA sample aliases AMZLL_BIN_001–AMZLL_BIN_050 and analysis accessions ERZ29768260–ERZ29768309 are included in Supplementary Table 7. MAG abundance profiles showed heterogeneous representation among lake-season groups, with notable contributions from Nitrospirae bacterium GWB2_47_37, Candidatus Nitrosotalea sp. TS, Trebonia kvetii, Euryarchaeota archaeon RBG_16_67_27 and other classified bins (Figure 7; Supplementary Figure 5).

antiSMASH parsing of the MAG/bin outputs identified BGC regions in 43 of 50 recovered MAGs, while MAG.2, MAG.5, MAG.20, MAG.32, MAG.44, MAG.47 and MAG.49 were retained in the manifest as having no BGC clusters identified (Supplementary Table 11). Across the parsed antiSMASH reports, we detected 146 BGC regions and 149 protoclusters. The product catalogue was dominated by terpene-precursor regions (44 regions across 32 MAGs), terpene regions (25 regions across 16 MAGs), NRPS-like regions (16 regions across nine MAGs), RRE-containing regions (10 regions across nine MAGs), NRPS regions (nine regions across two MAGs), T3PKS regions (seven regions across seven MAGs), betalactone regions (seven regions across six MAGs) and ranthipeptide/RiPP-like regions, with additional arylpolyene, ladderane, phosphonate, nucleoside, lanthipeptide, lasso peptide, CDPS, redox cofactor, triceptide, darobactin, hserlactone and NAPAA annotations (Supplementary Table 11).

Discussion

This study provides an integrated taxonomic, functional and genome-resolved assessment of microbial communities inhabiting Amazonian lateritic lakes developed over ferruginous canga. Across Violão, Amendoim, Três Irmãs and Três Irmãs Adjacent, Bacteria and Archaea dominated the classified fraction, yet more than half of the coding sequences remained unclassified. The detected lineages are broadly compatible with the diversity of freshwater lake microbiomes [44], whereas the large unclassified component supports the presence of microbial dark matter and poorly represented genomic diversity [45–47]. By linking these community profiles to curated metabolic markers, iron-gene annotations and 50 recovered MAGs, the study extends previous geological and limnological descriptions of the Carajás lakes into a genome-resolved microbial framework. Beyond taxonomic novelty, these sediment communities mediate organic-matter mineralization and nutrient availability, processes that may influence aquatic productivity and the belowground conditions experienced by endemic macrophytes [14,16,17].

Community differences were expressed primarily as lake-by-season contrasts rather than as a single, universal dry-versus-rainy response. This pattern suggests that basin morphology, sediment provenance, organic-matter supply, antecedent water levels and local redox structure jointly condition community assembly, instead of season acting as an isolated driver. That interpretation is consistent with the documented environmental heterogeneity of the Carajás upland lakes [8–16] and with ecological frameworks in which resistance and resilience depend on disturbance history and the availability of alternative metabolic niches [48]. Because the sampling design contains one dry and one rainy campaign, these associations should be viewed as spatially and seasonally structured observations rather than replicated estimates of a general seasonal effect.

Dry-season inter-lake contrasts included methanogenic, ammonia-oxidizing and ultrasmall archaeal lineages. A plausible interpretation is that hydrological drawdown and sediment exposure increase microscale redox heterogeneity, allowing anoxic refugia to persist within otherwise oxygen-exposed sediment. Methanogenesis can occur in spatially restricted microenvironments despite bulk oxygen exposure [49], while acidophilic ammonia oxidizers and Micrarchaeota-related organisms can tolerate oligotrophic or chemically stressful habitats [50,51]. *Methanoregula*ceae- and *Methanoregula*-related signals are also compatible with hydrogen-, carbon dioxide- or formate-linked methanogenic potential [52]. These observations do not demonstrate methane production, but they identify taxa and genomes that can be prioritized for porewater, transcript and flux measurements.

Rainy-season contrasts included anaerobic methanotroph-, Nitrosopumilales- and Nitrososphaera-related taxa, raising the hypothesis that methane and nitrogen transformations may be distributed across hydrologically reorganised sediment niches. Anaerobic methanotrophic archaea can limit methane escape from anoxic sediments [53], whereas Nitrosopumilales and Nitrososphaera include ammonia oxidizers that contribute to nitrification [54]. The occurrence of Nitrosopumilus-related genomes is additionally relevant because related representatives encode methylphosphonate biosynthesis pathways associated with aerobic methane formation [55,56]. The metagenomic data, however, neither establish simultaneous activity nor quantify process direction, and therefore support a set of testable mechanisms rather than a resolved methane budget. ANME lineages typically oxidize methane through reverse methanogenesis; their detection therefore provides a mechanistic basis for testing the hypothesis that anoxic microsites within ferruginous canga sediments act as a microbial methane filter, potentially limiting methane transfer to the water column and atmosphere, although methane-oxidation rates were not measured here [53]. Because seasonal hydrology influences oxygen penetration and sediment redox conditions, climate-driven changes in rainfall patterns and dry-period duration could alter the efficiency of this potential methane sink. The diversity and ordination analyses reinforce a heterogeneous ecosystem model. Rarefaction supported the retained sequencing depth, and NMDS showed partial overlap rather than discrete lake-season clusters. The domain-specific global RDA models were not significant (Bacteria: constrained $R^2 = 0.671$, adjusted $R^2 = 0.012$, pseudo-F = 1.02, permutation $P = 0.537$; Archaea: constrained $R^2 = 0.606$, adjusted $R^2 = -0.183$, pseudo-F = 0.77, permutation $P = 0.698$). Relationships with LOI, SiO_2, Al_2O_3, total sulfur, Cu and Pb should therefore be treated as exploratory covariation. Such caution is important because physicochemical heterogeneity can structure microbial communities [57], but the present ten-position integration does not provide evidence that any individual variable caused the observed taxonomic differences.

Functional reconstruction indicated a broad encoded repertoire spanning carbon fixation, respiratory complexes, methane metabolism, nitrogen transformations, sulfur assimilation, photosynthesis and heme or siroheme metabolism. This breadth is consistent with microorganisms serving as major engines of biogeochemical cycling [58,59] and with sediment communities in which methane, sulfur and nitrogen pathways coexist across redox gradients [60]. The coexistence of complete or near-complete modules does not imply that all pathways were active in the same sample or at the same time. Instead, the results indicate metabolic optionality across a heterogeneous community, to be resolved by expression-based measurements and geochemical rate assays.

Iron provides the central geochemical context for these functions. In canga sediments, iron minerals constitute both a lithological matrix and a reactive pool that can act as electron donor, electron acceptor and sorbent for phosphorus, organic matter and trace elements. Microbial Fe(II) oxidation and Fe(III) reduction can therefore connect mineral transformations to carbon, nitrogen, sulfur and methane cycling across shifting redox interfaces [61–63]. This framework explains why iron-associated genes should be interpreted together with organic-matter availability, oxygen penetration and mineral reactivity rather than as an isolated pathway inventory.

The FeGenie, iron-KO and MAG results are mutually consistent at the level of encoded potential. Iron-related markers occurred in MAGs affiliated with *Anaeromyxobacter*, Desulfobacterales or Deltaproteobacteria, Acidobacteria, Burkholderiales and *Nitrospira*les, lineages capable of occupying different positions along redox and organic-matter gradients. A plausible model is that flooding favours more reducing microhabitats, whereas sediment exposure increases oxygen penetration and iron re-precipitation; nevertheless, the present data do not measure iron speciation or reaction rates and cannot assign these processes to a particular season [61–63]. Direct microsensor, mineralogical and activity measurements are required to distinguish biotic iron turnover from abiotic reactions.

The external records place the Amazonian profiles within a broader spectrum of iron-rich ecosystems without converting heterogeneous studies into a controlled meta-analysis. Ferruginous Lake Matano and Lake Towuti provide tropical-lake analogues in which photoferrotrophy, methane cycling and organic-matter mineralization interact with iron-rich water columns and sediments [64–67]. Iron Mountain represents pyrite-driven acid mine drainage and a foundational genome-resolved view of an iron-rich community [68,69], whereas hydrothermal iron mats provide well-characterized examples of Zetaproteobacteria and Cyc2-linked iron oxidation [70,71]. Against that context, the Amazonian-positive directional contrasts emphasize methanogenesis, denitrification, dimethyl sulfoxide-linked respiration, formate dehydrogenase and selected iron-reduction or heme-linked markers, while external records more strongly represent phototrophy, methylotrophy, siderophore-related acquisition and oxidative iron handling (Supplementary Figures 68 and 69). These contrasts are useful for prioritizing hypotheses, but differences in habitat, sequencing depth, omics layer and annotation history preclude inferential claims of enrichment across studies.

The 50-MAG catalogue is an important contribution because it preserves candidates from both well-represented and poorly characterized lineages. High-completeness bins related to Nitrospirales, Euryarchaeota, cyanobacteria and Burkholderiales provide genomic anchors for nitrogen, carbon, photosynthetic and respiratory processes, while Bathyarchaeota-, Lokiarchaeota-, Methanoperedens-, Dehalococcoidia- and Aminicenantes-related MAGs expand the genomic representation of canga sediments. The complete quality, lineage, abundance and ENA accession information is retained in Supplementary Table 7; functional assignments to individual MAGs should remain provisional where genomes are incomplete or taxonomy is unresolved. The antiSMASH analysis provides additional insight into the secondary metabolic potential of the recovered MAGs. Terpene precursor, terpene, peptide-like, polyketide-like, betalactone, phosphonate and ladderane associated regions indicate chemically diverse biosynthetic capacity among the recovered MAGs [43]. These predictions are not evidence of metabolite production, but they define a reproducible set of targets for transcriptomics, metabolomics and comparative genome mining in ferruginous Amazonian sediments.

Several limitations delimit the inferences. Metagenomes measure encoded potential rather than activity; methane flux, nitrification, denitrification, sulfur reduction, iron reduction and iron oxidation were not measured directly. The environmental integration included ten positions and a non-significant global RDA, and the two sampling campaigns do not capture interannual variability. Cross-study contrasts combine different habitats, sequencing designs and omics layers. Future work should therefore pair broader spatial and temporal replication with porewater redox profiles, iron speciation, dissolved organic matter, stable-isotope incubations, metatranscriptomics, metaproteomics and direct greenhouse-gas or nutrient-transformation rates. In a broader ecological context, the breadth of taxonomic and functional diversity may contribute to ecosystem multifunctionality, because belowground biodiversity can support multiple coupled processes and resilience to environmental change [72]. Because these lake levels are controlled by the precipitation–evaporation balance, longer or more frequent low-water periods could alter sediment oxygen exposure and redox zonation, potentially redistributing methanogenic, nitrifying and iron-cycling guilds [8–10]. This is a testable projection rather than a demonstrated climate response. Paired sampling of *Isoetes* tissues, rhizosphere or adjacent sediments, porewater and mineral phases could additionally test whether microbial functions contribute to the contrasting habitat specificity of *I. cangae* and *I. serracarajensis* [16].

In summary, Amazonian lateritic canga lakes harbour taxonomically diverse and metabolically versatile sediment microbiomes with broad potential for carbon, methane, nitrogen, sulfur and iron transformations. The integrated atlas links community composition, exploratory environmental associations, curated marker profiles, iron-gene annotations and genome-resolved candidates while retaining the distinction between genomic potential and measured process. It establishes a reproducible baseline for testing how seasonal hydrology and ferruginous geochemistry organize microbial functions in tropical lateritic-lake ecosystems and identifies the measurements needed to move from encoded potential to quantified biogeochemical rates.

Supplementary material

Supplementary Tables 1–12 and Supplementary Figures 1–70 are provided as separate submission files, with complete legends in the supplementary-information document. Supplementary Table 1 consolidates the sequencing and assembly statistics, geographic coordinates and taxonomic source sheets used in the study. Supplementary Table 2 contains the taxonomic differential-abundance results; Supplementary Tables 3 and 9 contain

KEGG/KEMET module-completeness results for metagenomes and recovered MAGs, respectively; Supplementary Tables 4 and 5 contain curated KO-marker and pathway results; Supplementary Table 6 contains the complete annotation summary; Supplementary Table 7 contains MAG quality, taxonomy, abundance and ENA sample and analysis accessions; Supplementary Table 8 contains metadata and comparative results for the external iron-rich records; Supplementary Table 10 contains the general metrics and statistical summaries; and Supplementary Table 11 contains the antiSMASH MAG/bin biosynthetic-gene-cluster results. Supplementary Figures 1–13 report sequencing, assembly, diversity, KO-marker and metadata summaries; Supplementary Figures 14–25 provide GTDB taxonomic profiles separated by taxonomic level and omics layer; Supplementary Figures 26–31 provide taxonomic-overlap, workflow, coordinate and common-taxa summaries; Supplementary Figures 32–36 provide biogeochemical-cycle and cross-study marker contrasts; Supplementary Figures 37–42 provide MAG, metagenome and external-record KEGG-module, differential-taxon and lake-season KO summaries; Supplementary Figures 43–66 provide domain-separated Bacteria and Archaea barplots and log10(relative abundance [%] + 1) heatmaps at phylum, class, order, family, genus and species levels; Supplementary Figure 67 provides the combined lagoon and external iron-rich module-completeness heatmap; and Supplementary Figures 68 and 69 provide the biogeochemical and iron-associated KO directional contrasts. A separate Supplementary Methods document provides the complete computational and figure-reproduction protocol, including script-specific inputs, processing steps, commands, outputs and validation procedures.

Figure files are supplied in PNG, PDF and SVG formats, and the exact plotted values are provided in the corresponding supplementary data tables.

Data availability

Sequencing data were reported in the supplied manuscript as deposited in the GOLD database under project application number Gp0646171. Source data, processed figure tables and validation files are included in this submission package.

The raw metagenomic sequencing data are publicly available in the European Nucleotide Archive under study accession ERP137391 and run accessions ERR9980190–ERR9980209. All annotated MAGs are publicly available in the BV-BRC workspace at https://www.bv-brc.org/workspace/mattoslmp@patricbrc.org/Lakes-Canga/metagenomas [38]. The recovered MAGs are also deposited as ENA analyses under accessions ERZ29768260–ERZ29768309, with MAG-to-accession mapping provided in Supplementary Table 7.

Code availability

Analysis scripts, figure-generation code, the distributable reader package, source data tables, supplementary material, and the Python/SLURM workflows for metagenome assembly and Kaiju processing are available in the project repository at https://github.com/mattoslmp/GangaMetaG-IronMetagenomicAtlas. The Supplementary Methods provide a standalone script-by-script protocol listing the required inputs, implemented transformations, commands, outputs, software dependencies and random-seed policies for all canonical main and supplementary figures. The Kaiju parsing workflow referenced in the manuscript is also available at https://github.com/mattoslmp/kaiju2-data-parser.

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Author contributions

Leandro de Mattos Pereira: Conceptualization; Methodology; Software; Formal analysis; Investigation; Validation; Data curation; Visualization; Writing – original draft; Writing – review and editing. Gisele Lopes Nunes: Conceptualization; Methodology; Investigation; Supervision; Project administration; Funding acquisition, Validation; Writing – review and editing.

[AUTHOR TO CONFIRM] Add author-contribution statements using the journal's required taxonomy. Suggested structure: conceptualization, field sampling, laboratory processing, bioinformatics, data curation, formal analysis, visualization, writing—original draft, writing—review and editing, supervision and funding acquisition.

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Conflicts of interest

The authors declare no conflicts of interest.

Ethics approval and permits

[AUTHOR TO CONFIRM] Add collection permit, institutional authorization or statement confirming that no ethics approval was required for sediment metagenomic sampling.

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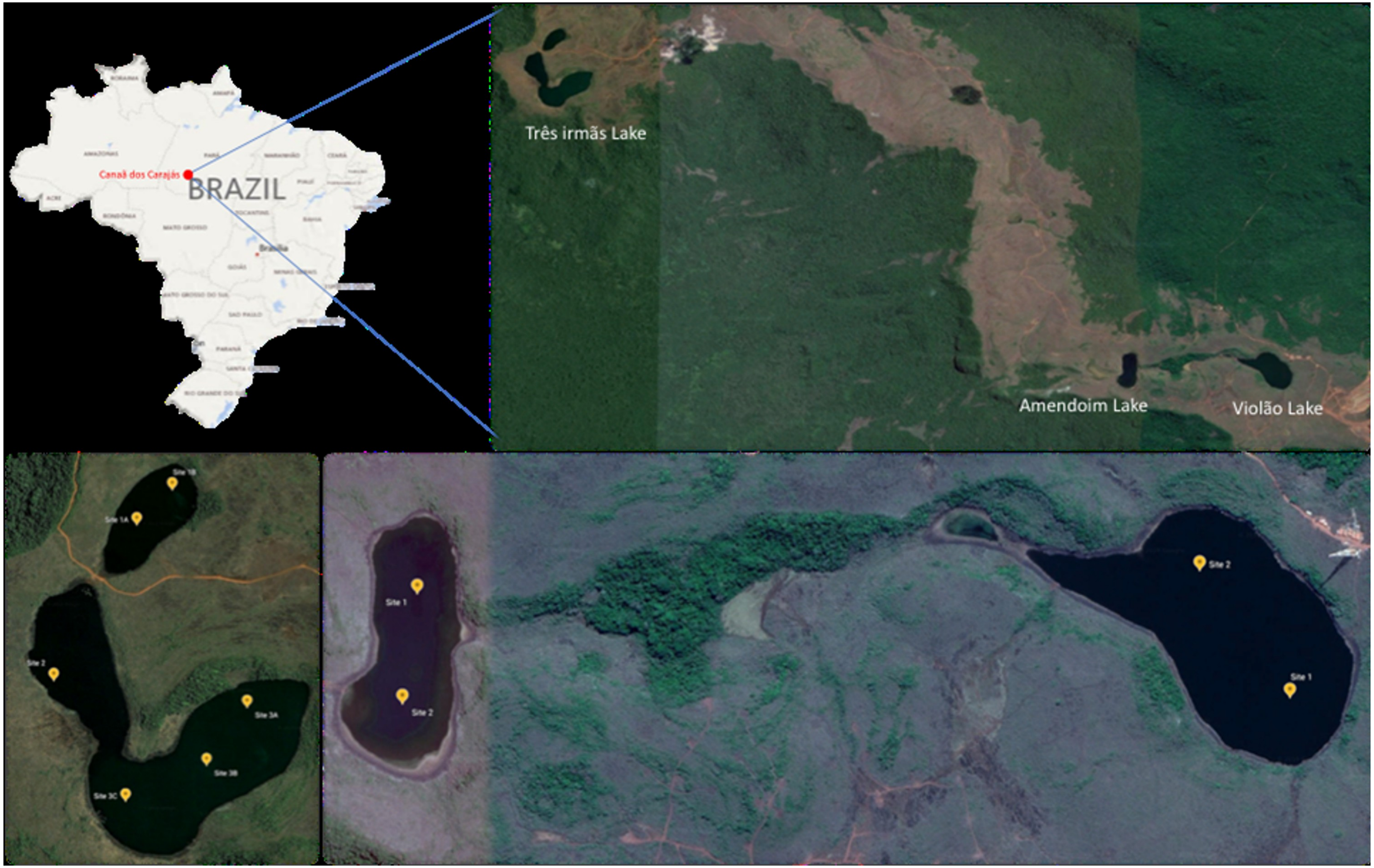

Figure 1. Geographic location and sampling sites of the Amazonian lateritic lakes analysed in this study. Amendoim, Violão and Três Irmãs lakes are located on ferruginous canga plateaus in Serra dos Carajás, Brazil. Sampling points correspond to dry- and rainy-season sediment collections.

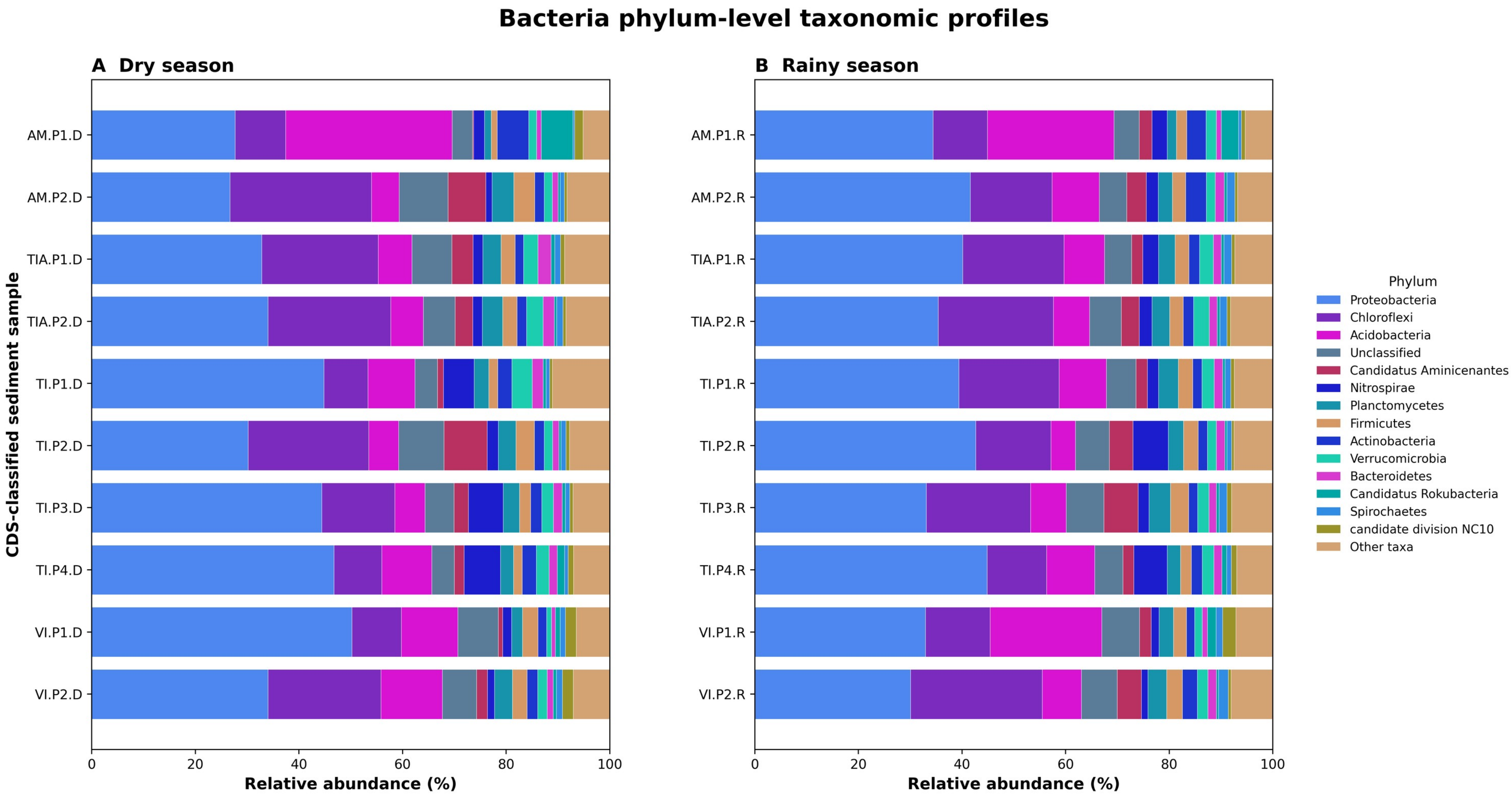

Figure 2. CDS-based phylum-level taxonomic profiles of Bacteria in Amazonian lateritic-lake sediment metagenomes. Panels A and B show dry- and rainy-season samples, respectively. Stacked bars represent relative abundance (%).

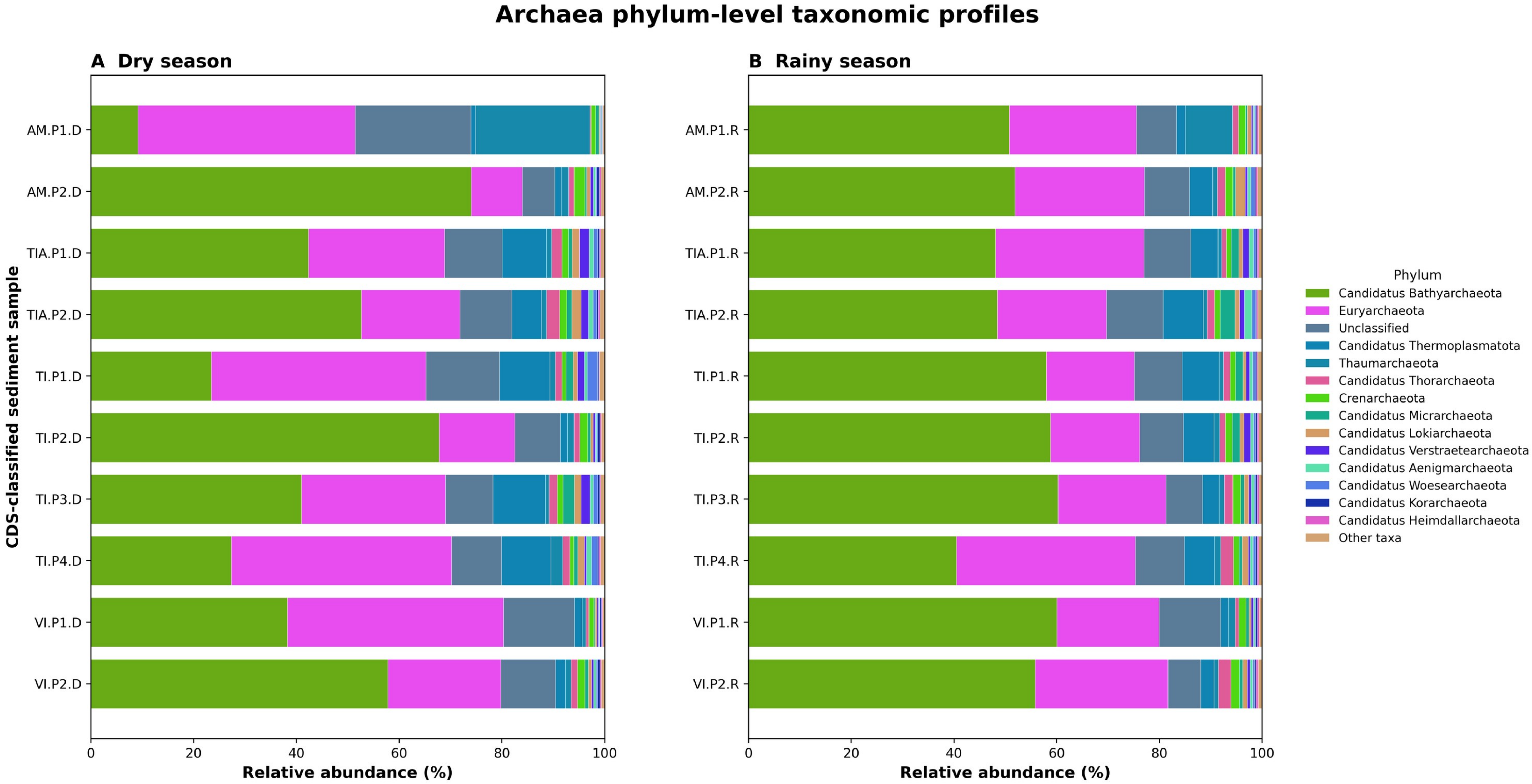

Figure 3. CDS-based phylum-level taxonomic profiles of Archaea in Amazonian lateritic-lake sediment metagenomes. Panels A and B show dry- and rainy-season samples, respectively. Stacked bars represent relative abundance (%).

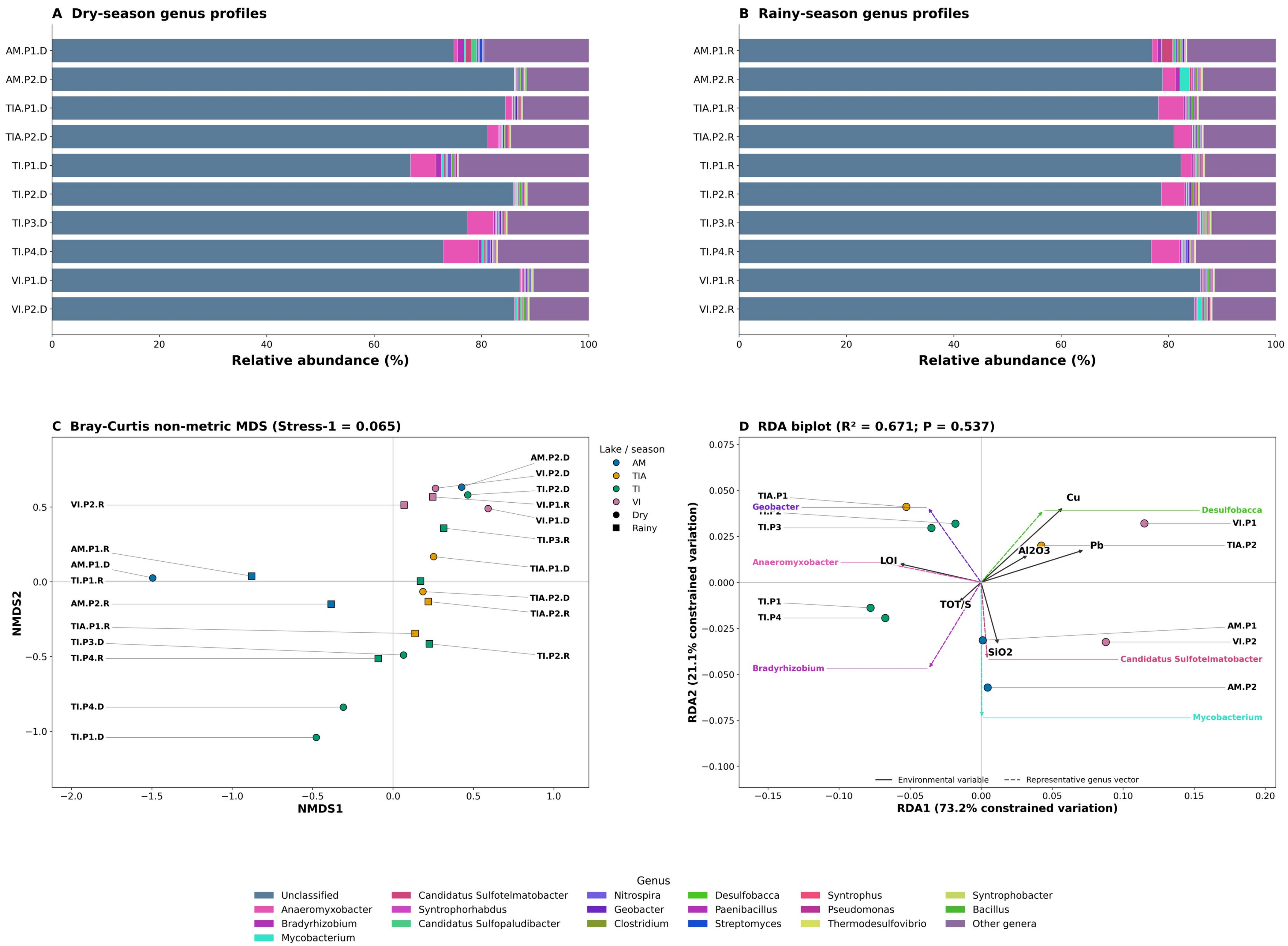

Figure 4. Bacterial genus-level taxonomic profiles and ordination. (A) Dry-season and (B) rainy-season genus-level relative-abundance profiles. (C) True two-dimensional non-metric MDS of all 20 metagenomes using Bray–Curtis dissimilarity of square-root-transformed relative proportions (normalized Stress-1 = 0.065; 20 initializations; seed 42). (D) Exploratory RDA of ten physicochemical sampling positions. Dry and rainy metagenomes were pooled only for the RDA to match the ten independent position-level chemistry measurements; all ten included positions are drawn and labelled. Predictors were LOI, SiO2, Al2O3, total sulfur, Cu and Pb.

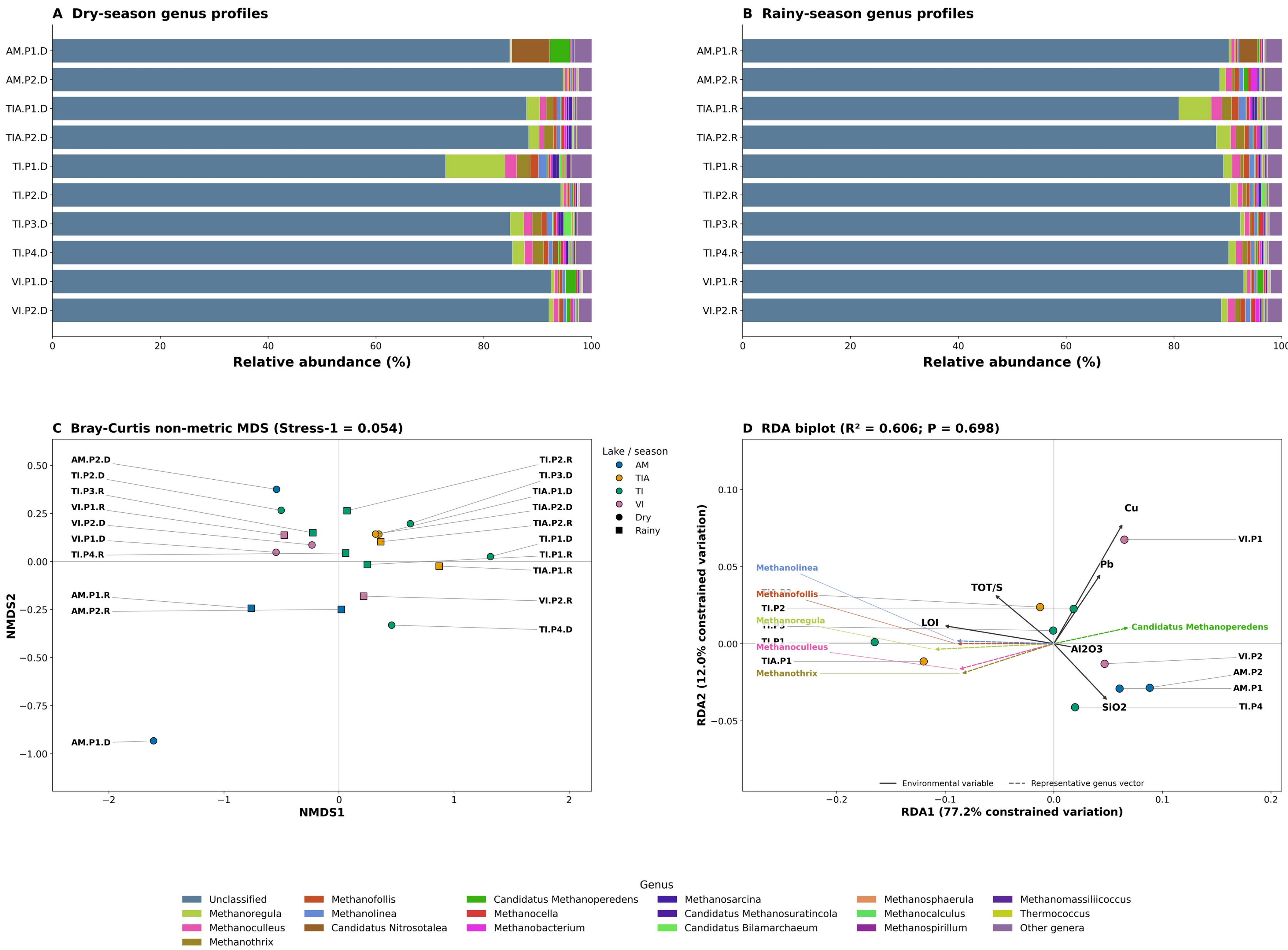

Figure 5. Archaeal genus-level taxonomic profiles and ordination. (A) Dry-season and (B) rainy-season genus-level relative-abundance profiles. (C) True two-dimensional non-metric MDS of all 20 metagenomes using Bray–Curtis dissimilarity of square-root-transformed relative proportions (normalized Stress-1 = 0.054; 20 initializations; seed 42). (D) Exploratory RDA of ten physicochemical sampling positions. Dry and rainy metagenomes were pooled only for the RDA to match the ten independent position-level chemistry measurements; all ten included positions are drawn and labelled. Predictors were LOI, SiO2, Al2O3, total sulfur, Cu and Pb.

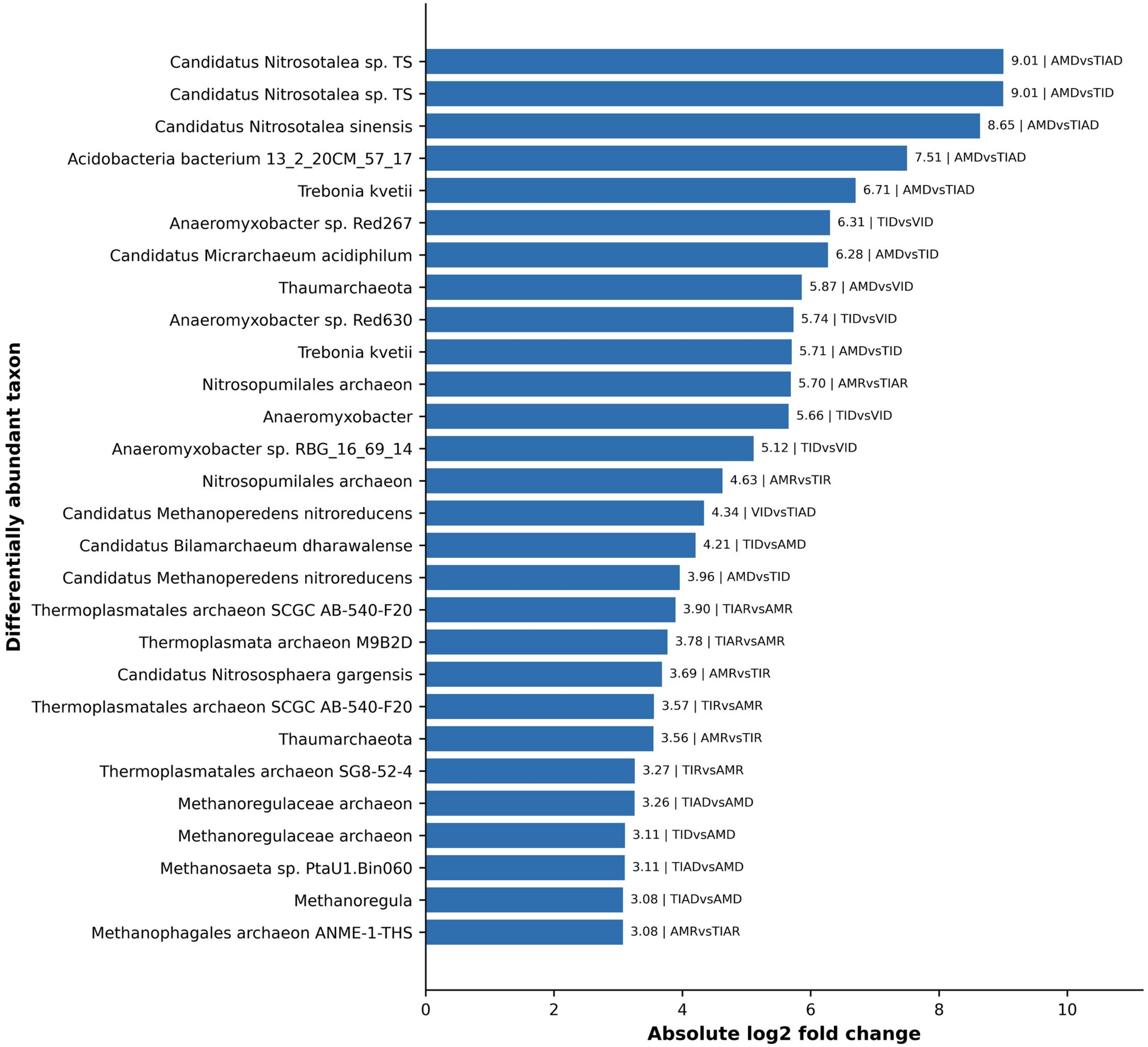

Figure 6. Differential abundance of bacterial and archaeal taxa across lake-season comparisons. Bars show directional log2 fold-change values, and labels identify the corresponding comparison.

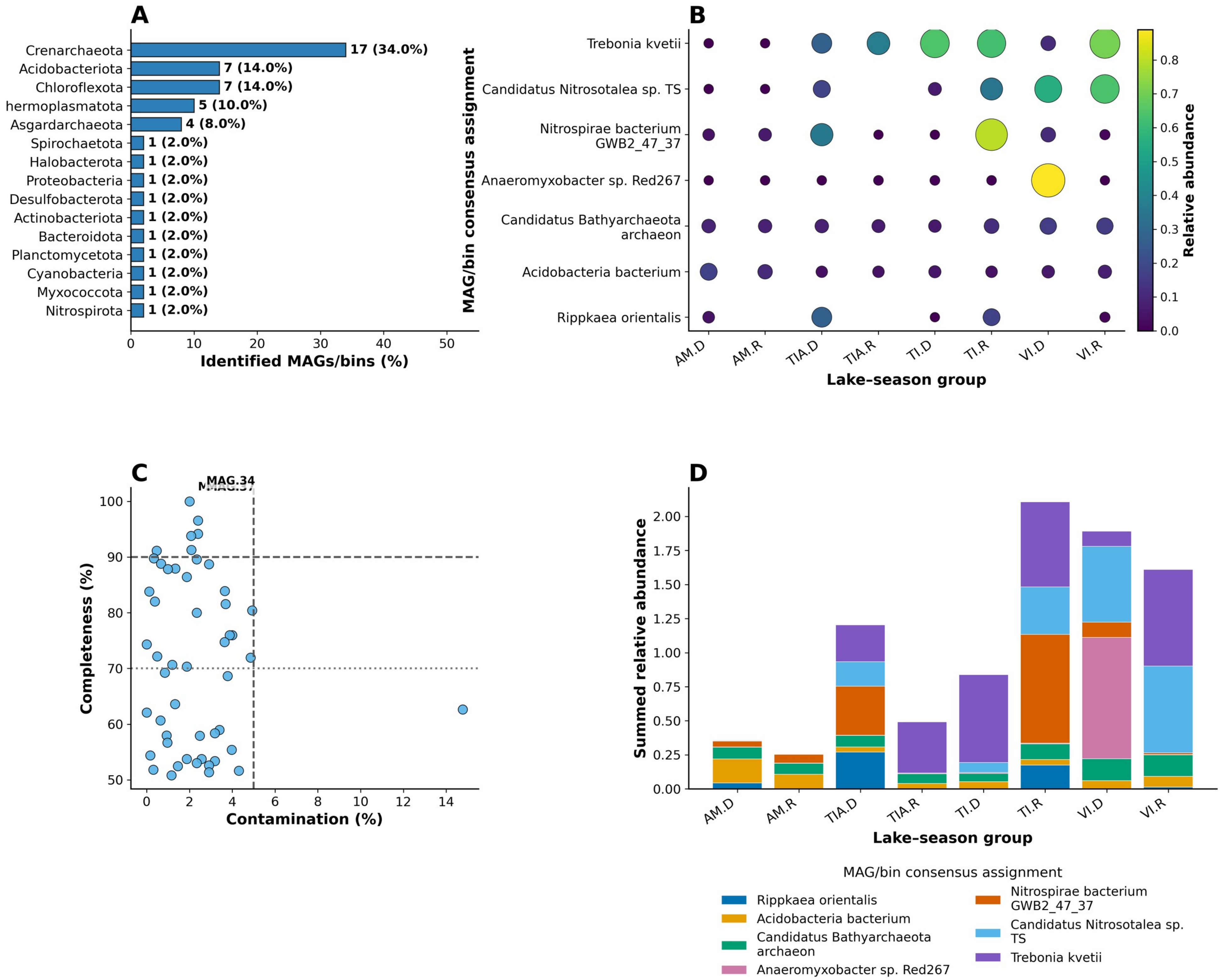

Figure 7. MAG quality, classification and abundance across lake-season groups. (A) Distribution of recovered MAGs/bins among GTDB/CheckM phylum assignments. (B) Relative-abundance bubble plot for the seven most abundant consensus assignments; bubble area and colour scale with abundance. (C) CheckM completeness and contamination, with 90%/70% completeness and 5% contamination guides. (D) Summed relative abundance of the same assignments by lake-season group.

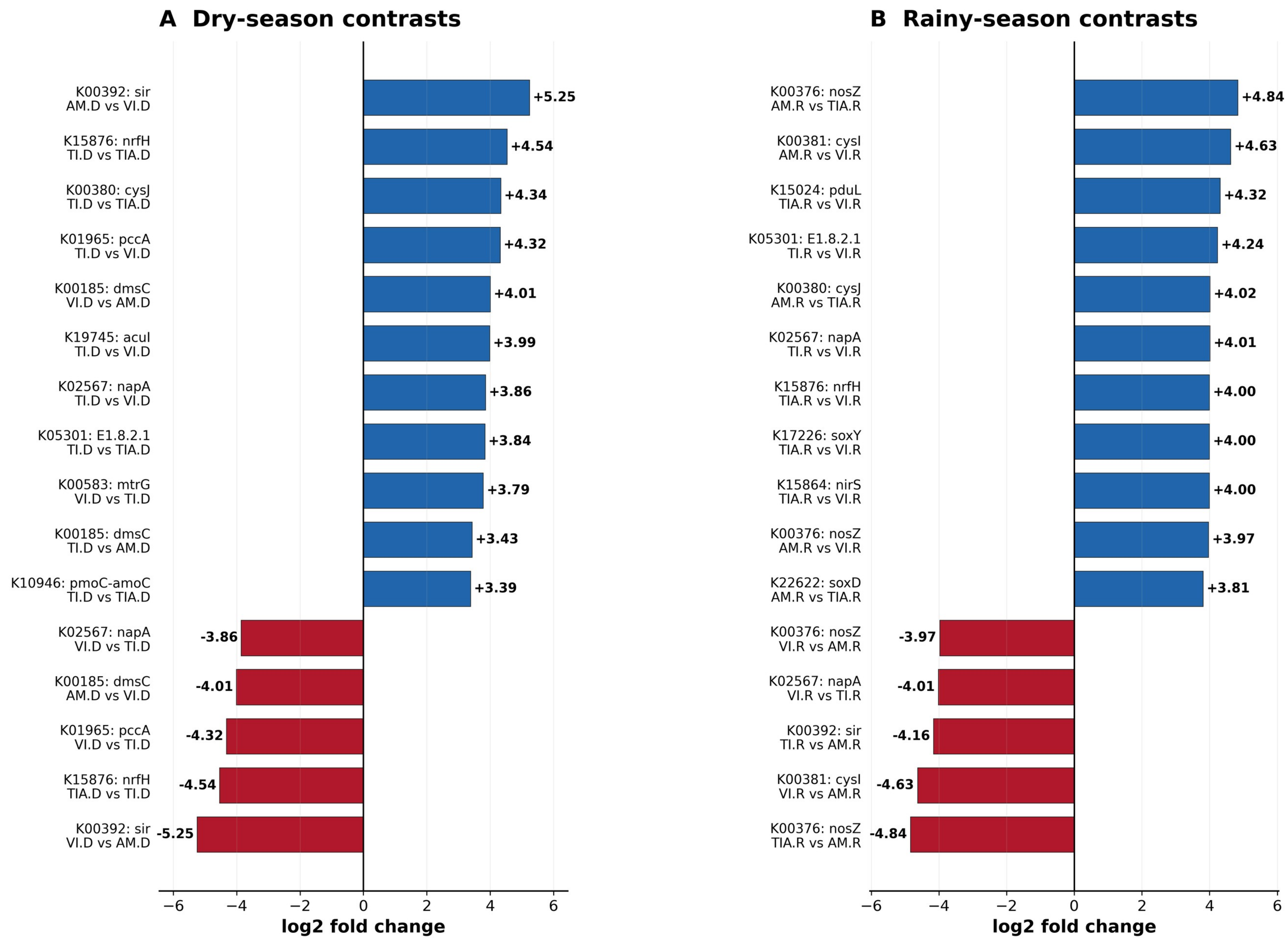

Figure 8. Differential abundance of KEGG Orthology biomarkers in dry- and rainy-season pairwise lake contrasts. Each bar is labelled with the KO and the actual directional comparison derived from Supplementary Table 5. AM, TI, TIA and VI denote Amendoim, Três Irmãs, Três Irmãs Adjacent and Violão, respectively; D and R denote dry and rainy periods. For a comparison A vs B, positive log2 fold-change values indicate higher abundance in A and negative values indicate higher abundance in B. Bars show the 16 largest absolute contrasts in each season from the reported results; statistical significance and method fields are provided in Supplementary Table 5.